Bayesian Framework for Canonical factorization of multiblock data Bayesian Framework for Canonical factorization of multiblock data

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1 Introduction

Data sets structured into blocks of variables measured on the same set of observations are frequently encountered in practice. These blocks may consist of quantitative variables originating from distinct measurement sources or correspond to thematic groupings defined by the user. For instance, in omics studies, blocks typically comprise variables derived from different analytical techniques (e.g., proteomics, metabolomics). In psychology, multiblock data structures commonly arise when multiple sets of variables are collected on the same individuals to capture different psychological domains. For example, distinct blocks may include self-report questionnaire scores assessing personality traits, behavioral task performance measures, and neurophysiological indicators such as electroencephalography features.

The four expected objectives of exploratory multiblock data analysis are: (i) to perform a joint dimensionality reduction, (ii) to examine relationships among blocks, (iii) to investigate associations among variables both within and across blocks, and (iv) to capture within-block variability in order to reveal shared and block-specific structures in a common latent space. An unified algebraic framework based on factorization strategies, denoted *canonical factorisation*, was proposed by [Bougeard et al., 2024] for multiblock methods to achieve these objectives. In practice, canonical factorisation provides a structured framework for analysing data variability at multiple levels: differences between and within blocks, differences between observations by projecting them onto a common subspace. It also facilitates the harmonisation of outputs produced by different methods and supports a more objective criterion-based selection among competing approaches. The four expected objectives of exploratory multiblock data analysis are: (i) to perform a joint dimensionality reduction, (ii) to examine relationships among blocks, (iii) to investigate associations among variables both within and across blocks, and (iv) to capture within-block variability in order to reveal shared and block-specific structures in a common latent space. An unified algebraic framework based on factorization strategies, denoted *canonical factorisation*, was proposed by [Bougeard et al., 2024] for multiblock methods to achieve these objectives. In practice, canonical factorisation provides a structured framework for analysing data variability at multiple levels: differences between and within blocks, differences between observations by projecting them onto a common subspace. It also facilitates the harmonisation of outputs produced by different methods and supports a more objective criterion-based selection among competing approaches.

2 Canonical factorisation

Notations

- K blocks $(\mathbf{X}_1, \ldots, \mathbf{X}_K)$ respectively of $p_1, \ldots, p_K$ variables on N individuals,

- $\mathbf{X} = [\mathbf{X}_1|\dots|\mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ ($P = \sum_{k=1}^{K} p_k$) be the concatenation of these K-blocks,
- $\mathbf{U}^\intercal$ designates the transposed matrix of $\mathbf{U}$,
- $\Sigma_k = diag(\sigma_k^{(1)}, ..., \sigma_k^{(R)})$ are diagonal matrices
- $\|.\|$ correspond to the Frobenius norm,
- $\mathcal{V}_{N,R} = \{\mathbf{W} \in \mathcal{M}_{N,R}(\mathbb{R}) : \mathbf{W}^\intercal\mathbf{W} = \mathbf{I}_R\}$ denotes the Stiefel manifold.

Consider K colmumn blocks $(\mathbf{X}_1, \dots, \mathbf{X}_K)$ respectively in $\mathcal{M}_{N,p_1}(\mathbb{R}), ..., \mathcal{M}_{N,p_K}(\mathbb{R}))$. Let $\mathbf{X} = [\mathbf{X}_1|\dots|\mathbf{X}_K] \in \mathcal{M}_{N,P}(\mathbb{R})$ ($P = \sum_{k=1}^{K} p_k$) be the concatenation of these K-blocks. Without loss of generality, the blocks are assumed to be column-centered.

[Bougeard et al., 2024] demonstrated that the $(K+1)-$blocks $(\mathbf{X}_1, \dots, \mathbf{X}_K, \mathbf{X})$ can be factorized in dimension $R = rank(\mathbf{X})$ as follows

$$\begin{cases} \mathbf{X} = \mathbf{U}\Sigma\mathbf{V}^\intercal \\ \mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal \quad \text{for} \quad k = (1, \dots, K) \\ \text{with the constraints} \quad \mathbf{U} \in \mathcal{V}_{N,R} \quad \text{and} \quad \|\mathbf{v}_k^{(1)}\| = \dots = \|\mathbf{v}_k^{(R)}\| = 1 \quad \text{for} \quad k = (1, \dots, K), \end{cases} \tag{1}$$

where

- $\mathbf{U} \in \mathcal{V}_{N,R}$ is the matrix of common components,
- $\mathbf{V}_k \in \mathcal{M}_{p_k,R}(\mathbb{R})$ are the standardized block-loadings, and $\mathbf{V} = [\mathbf{V}_1|...|\mathbf{V}_K] \in \mathcal{M}_{P,R}(\mathbb{R})$,
- $\Sigma \in \mathcal{M}_{R,R}(\mathbb{R})$ the diagonal scaling matrices with property $\Sigma^2 = \sum_{k=1}^{K} \Sigma_k^2$ that relate common and block-analyses.

Without loss of generality, we can assume $\sigma_k^{(r)} \geq 0$ for all k, r, since a joint sign change of $\sigma_k^{(r)}$ and $\mathbf{v}_k^{(r)}$ leaves the model and the constraints unchanged. This factorization is *the canonical factorization*. Each multiblock method derives a common component $\mathbf{U}$ using a different optimization criterion. Projecting the data onto these common components enables their canonical factorization (for details see [Bougeard et al., 2024]). For all the multiblock methods, once $\mathbf{U}$ is fixed, the decomposition $\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$ of each block is unique (up to sign). The methods differ in how the axes $\mathbf{u}^{(1)}, \dots, \mathbf{u}^{(R)}$ themselves are determined. When the block loadings are only normalized ($\|\mathbf{v}_k^{(r)}\| = 1$), the optimal loadings are $\mathbf{v}_k^{(r)} = \mathbf{X}_k^\intercal\mathbf{u}^{(r)}/\|\mathbf{X}_k^\intercal\mathbf{u}^{(r)}\|$ and the least-squares criterion reduces to

$$\sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}\|^2 - \|\mathbf{X}^\intercal\mathbf{U}\|^2,$$

which is invariant under any rotation $\mathbf{U} \mapsto \mathbf{U}\mathbf{R}$: only span($\mathbf{U}$) is determined by the data, and the axes have to be fixed by an external convention, usually the SVD of $\mathbf{X}$. This convention becomes unstable when the singular values of $\mathbf{X}$ are close. When $\mathbf{V}_k \in \mathcal{V}_{p_k,R}$ for all k, the criterion is no longer rotation invariant, and the axes of $\mathbf{U}$ are identified (up to permutation and sign) by the block-specific profiles $\Sigma_1, \dots, \Sigma_K$, as long as these profiles differ across blocks.

For $R < rank(\mathbf{X})$, the objective is to determine the optimal approximation of the form given in (1) by minimizing an ordinary least square criterion.

2.1 Canonical Factorisation under orthogonal block-loadings constraint

Providing a solution for canonical factorization means minimizing

$$\sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X} - \mathbf{U}\Sigma\mathbf{V}^\intercal\|^2, \tag{2}$$

with $\mathbf{U} \in \mathcal{V}_{N,R}$, $\mathbf{V}_k \in \mathcal{V}_{p_k,N}$ for $k = 1, \ldots, K$. Chessel and Hanafi [Chessel and Hanafi, 1996] proposed a sequential solution, known as Multiple Co-Inertia Analysis (MCOA): the dimensions are computed one at a time, each from the blocks deflated with respect to the previous block loadings. This procedure yields orthonormal matrices $\mathbf{U}$ and $\mathbf{V}_1, \ldots, \mathbf{V}_K$, and each stage is optimal in the least-squares sense given the previous ones. The resulting solution is nevertheless suboptimal with respect to (2), since each dimension is chosen without taking the next ones into account, so that the R dimensions are not fitted simultaneously. Optimal Multiple COinertia Ananlysis (OMCOA) is proposed as a simultaneous solution of (2) under the constraints $\mathbf{U} \in \mathcal{V}_{N,R}$ and $\mathbf{V}_k \in \mathcal{V}_{N,R}$ $(k = 1, \ldots, K)$. It is computed by an alternating least squares (ALS) algorithm, in which each step minimizes (2) with respect to one set of parameters, the others being fixed:

- **Update of U:** $\mathbf{U} = \mathbf{A}\mathbf{B}^\intercal$, where $\mathbf{X}\mathbf{H}^\intercal = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal$ is the SVD of $\mathbf{X}\mathbf{H}^\intercal$, with $\mathbf{H} = [\Sigma_1\mathbf{V}_1^\intercal | \cdots | \Sigma_K\mathbf{V}_K^\intercal]$ (orthogonal Procrustes problem);
- **Update of $\mathbf{V}_k$:** $\mathbf{V}_k = \mathbf{A}_k\mathbf{B}_k^\intercal$, where $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k = \mathbf{A}_k\mathbf{D}_k\mathbf{B}_k^\intercal$ is the SVD of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$, for $k = 1, \ldots, K$;
- **Update of Σ_k:** $\Sigma_k = \mathrm{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$, for $k = 1, \ldots, K$.

These steps are iterated until the decrease of (2) falls below a given tolerance. Since each step cannot increase the criterion, which is bounded below, the sequence of criterion values converges; the limit may however be a local minimum, so the algorithm is started from the MCOA solution and from several random initializations, and the solution with the lowest criterion is retained.

3 Bayesian framework for Canonical Factorisation

A probabilistic counterpart of (2) is obtained by assuming a Gaussian noise:

$$\mathbf{X}_k = \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal + \mathbf{E}_k, \qquad \mathbf{E}_k \text{ with i.i.d. entries } \mathcal{N}(0, \tau_k^{-1}), \qquad k = 1, \ldots, K, \tag{3}$$

where τ_k is the noise precision of block k. Since the blocks are centered, the latent components are constrained to be centered as well: $\mathbf{U} \in \mathcal{V}_{N,R}$ with $\mathbf{1}_N^\intercal\mathbf{U} = 0$.

Priors. We use the following independent priors:

- $\mathbf{U}$ and $\mathbf{V}_1, \ldots, \mathbf{V}_K$ are uniformly distributed on their Stiefel manifolds (non-informative);
- $\sigma_k^{(h)} \sim \mathcal{N}^+(0, \lambda^{-1})$, a half-normal distribution with a small precision λ (e.g. $\lambda = 10^{-2}$); the positivity constraint removes the sign indeterminacy between $\sigma_k^{(h)}$ and $\mathbf{v}_k^{(h)}$;
- $\tau_k \sim \mathrm{Gamma}(a_0, b_0)$ with small hyperparameters (e.g. $a_0 = b_0 = 10^{-3}$).

3.1 Full conditional distributions

Likelihood. Because $\mathbf{U}$ and $\mathbf{V}_k$ have orthonormal columns and Σ_k is diagonal,

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\,\mathrm{trace}(\Sigma_k\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}) + \sum_{h=1}^{H}(\sigma_k^{(h)})^2. \tag{4}$$

Since the blocks are centered, each $\mathbf{X}_k$ carries $(N-1)p_k$ degrees of freedom, and the log-likelihood reads

$$\log p(\mathbf{X}_1,\dots,\mathbf{X}_K \mid \cdot) = \sum_{k=1}^{K}\Big[\tfrac{(N-1)p_k}{2}\log\tau_k - \tfrac{\tau_k}{2}\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2\Big] + \text{const.}$$

By (4), $\mathbf{U}$ and $\mathbf{V}_k$ only appear in the linear term $\mathrm{trace}(\Sigma_k\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U})$. This is the key property: all the full conditionals belong to standard families.

Matrix von Mises–Fisher distribution. A random matrix $\mathbf{W} \in \mathcal{V}_{m,H}$ follows the matrix von Mises–Fisher distribution $\mathrm{vMF}(\mathbf{F})$, with parameter $\mathbf{F} \in \mathcal{M}_{m,H}(\mathbb{R})$, if its density with respect to the uniform measure on $\mathcal{V}_{m,H}$ is proportional to $\exp\{\mathrm{trace}(\mathbf{F}^\intercal\mathbf{W})\}$ [Hoff, 2009]. If $\mathbf{F} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal$ is the (thin) SVD of $\mathbf{F}$, the mode of $\mathrm{vMF}(\mathbf{F})$ is the polar factor $\mathbf{A}\mathbf{B}^\intercal$, and the distribution concentrates around it as the singular values of $\mathbf{F}$ increase.

Proposition 1 (Full conditionals)**.** *Under model* (3) *and the above priors:*

1. ***Common space.*** *Let $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ be an orthonormal basis of $\mathbf{1}_N^\perp$ and write $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$ with $\tilde{\mathbf{U}} \in \mathcal{V}_{N-1,H}$. Then*

$$\tilde{\mathbf{U}} \mid \cdot \;\sim\; \mathrm{vMF}(\mathbf{Q}^\intercal\mathbf{F}_{\mathbf{U}}), \qquad \mathbf{F}_{\mathbf{U}} = \sum_{k=1}^{K}\tau_k\,\mathbf{X}_k\mathbf{V}_k\Sigma_k.$$

2. ***Block loadings.*** *For $k = 1,\dots,K$,*

$$\mathbf{V}_k \mid \cdot \;\sim\; \mathrm{vMF}(\mathbf{F}_{\mathbf{V}_k}), \qquad \mathbf{F}_{\mathbf{V}_k} = \tau_k\,\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k.$$

3. ***Block weights.*** *For $k = 1,\dots,K$ and $h = 1,\dots,H$, let $d_k^{(h)} = \mathbf{u}^{(h)\intercal}\mathbf{X}_k\mathbf{v}_k^{(h)}$. Then*

$$\sigma_k^{(h)} \mid \cdot \;\sim\; \mathcal{N}^+\Big(\frac{\tau_k\,d_k^{(h)}}{\tau_k+\lambda},\; \frac{1}{\tau_k+\lambda}\Big),$$

   a normal distribution truncated to $(0,+\infty)$.

4. ***Noise precisions.*** *For $k = 1,\dots,K$,*

$$\tau_k \mid \cdot \;\sim\; \mathrm{Gamma}\Big(a_0 + \frac{(N-1)p_k}{2},\; b_0 + \frac{1}{2}\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2\Big).$$

We can deduce the following algorithm: We can deduce the following algorithm:

Algorithm 1 Gibbs sampler for OMCOA

Require: centered blocks $\mathbf{X}_1, \dots, \mathbf{X}_K$; dimension H; number of iterations T; burn-in T_0; hyperparameters a_0, b_0, λ

Initialization: $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$, $\mathbf{V}_k$, Σ_k from OMCOA (or MCOA, or at random); $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
for $t = 1, \dots, T$ **do**
  $\mathbf{F}_\mathbf{U} \leftarrow \sum_k \tau_k \mathbf{X}_k\mathbf{V}_k\Sigma_k$; update $\tilde{\mathbf{U}}$ by one column-wise sweep of vMF($\mathbf{Q}^\intercal\mathbf{F}_\mathbf{U}$); $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  for $k = 1, \dots, K$ **do**
    $\mathbf{F}_{\mathbf{V}_k} \leftarrow \tau_k \mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$; update $\mathbf{V}_k$ by one column-wise sweep of vMF($\mathbf{F}_{\mathbf{V}_k}$)
    for $h = 1, \dots, H$ **do**
      $d_k^{(h)} \leftarrow \mathbf{u}^{(h)\intercal}\mathbf{X}_k\mathbf{v}_k^{(h)}$; draw $\sigma_k^{(h)} \sim \mathcal{N}^+\big(\tau_k d_k^{(h)}/(\tau_k+\lambda),\ 1/(\tau_k+\lambda)\big)$
    end for
    draw $\tau_k \sim \text{Gamma}\big(a_0 + (N-1)p_k/2,\ b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$
  end for
  if $t > T_0$ **then** store $\big(\mathbf{U}, \mathbf{V}_1, \dots, \mathbf{V}_K, \Sigma_1, \dots, \Sigma_K, \tau_1, \dots, \tau_K\big)$
  end if
end for
Post-processing: align each stored draw on a reference solution (Section 3.1)
return the aligned draws

Post-processing: label switching and signs The posterior distribution is invariant under any permutation π of the H dimensions applied jointly to $\mathbf{U}$, $\mathbf{V}_1, \dots, \mathbf{V}_K$ and $\Sigma_1, \dots, \Sigma_K$, and under a joint change of sign of $\mathbf{u}^{(h)}$ and $\mathbf{v}_1^{(h)}, \dots, \mathbf{v}_K^{(h)}$. Posterior summaries are therefore computed after aligning each draw on a reference $\mathbf{U}_{\text{ref}}$ (in practice the OMCOA solution). For each draw, with $\mathbf{C} = \mathbf{U}_{\text{ref}}^\intercal\mathbf{U}$,

$$\hat{\pi} = \operatorname*{arg\,max}_{\pi \in \mathfrak{S}_H} \sum_{h=1}^{H} |C_{h,\pi(h)}|, \qquad s_h = \text{sign}\big(C_{h,\hat{\pi}(h)}\big),$$

and the columns of $\mathbf{U}$ and $\mathbf{V}_k$ are reordered by $\hat{\pi}$ and multiplied by s_h, while the diagonal of Σ_k is reordered by $\hat{\pi}$ (it stays positive). Posterior means and credible intervals of $\sigma_k^{(h)}$ are then obtained from the empirical distribution of the aligned draws.

Remark 1 (OMCOA as iterated conditional modes)**.** *The modes of the full conditionals of Proposition 1 are*

$$\mathbf{U} = \mathbf{A}\mathbf{B}^\intercal \text{ with } \mathbf{F}_\mathbf{U} = \mathbf{A}\mathbf{D}\mathbf{B}^\intercal, \qquad \mathbf{V}_k = \mathbf{A}_k\mathbf{B}_k^\intercal \text{ with } \mathbf{F}_{\mathbf{V}_k} = \mathbf{A}_k\mathbf{D}_k\mathbf{B}_k^\intercal, \qquad \sigma_k^{(h)} = \frac{\tau_k}{\tau_k+\lambda}\, d_k^{(h)}.$$

When $\lambda \to 0$ and the precisions are equal ($\tau_1 = \dots = \tau_K$), these are exactly the OMCOA updates: Procrustes rotation of $\mathbf{X}\mathbf{H}^\intercal$ for $\mathbf{U}$ (with $\mathbf{H} = [\Sigma_1\mathbf{V}_1^\intercal | \dots | \Sigma_K\mathbf{V}_K^\intercal]$), Procrustes rotation of $\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k$ for $\mathbf{V}_k$, and $\Sigma_k = \text{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$. OMCOA is therefore the iterated conditional modes (ICM) algorithm [Besag, 1986] associated with this posterior, and the Gibbs sampler is its stochastic counterpart.

Remark 2 (Block weighting)**.** *With block-specific precisions, $\mathbf{F}_\mathbf{U} = \sum_k \tau_k\mathbf{X}_k\mathbf{V}_k\Sigma_k$ weights each block by its estimated noise precision, so that noisier blocks contribute less to the common space. To reproduce exactly the OMCOA criterion* (2)*, a common precision $\tau_1 = \dots = \tau_K = \tau$ can be used, with* (2)*, a common precision $\tau_1 = \dots = \tau_K = \tau$ can be used, with $\tau \mid \cdot \sim \text{Gamma}\big(a_0 + \sum_k (N-1)p_k/2,\ b_0 + \sum_k \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$.*

Remark 3 (What the sampler adds)**.** *Besides point estimates close to the OMCOA solution, the sampler provides the posterior distribution of the block weights $\sigma_k^{(h)}$ and of the noise levels τ_k^{-1}. Credible intervals on $\sigma_k^{(h)}$ can be used to assess whether a dimension is common to all blocks, shared by some of them, or specific to one block.*

Remark 4 (Practical aspects)**.** *Starting from the OMCOA solution shortens the burn-in. The column-wise sweeps make successive draws of $\mathbf{U}$ and $\mathbf{V}_k$ correlated, so convergence should be checked on trace plots of the $\sigma_k^{(h)}$. Because the posterior is strongly concentrated when the noise is small, the chain rarely jumps between the $H!$ equivalent modes; the alignment step remains necessary nonetheless, in particular when several dimensions have close weights.*

3.2 sparse version

In OMCOA, each block is approximated as $\mathbf{X}_k \approx \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal$, where the diagonal matrix $\Sigma_k = \mathrm{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$ measures how much each latent dimension $\mathbf{u}^{(r)}$ contributes to block k. The structure of the blocks can be read from the pattern of zeros of $\Sigma_1, \ldots, \Sigma_K$: dimension r is

- *U-common* if $\sigma_k^{(r)} > 0$ for all k;
- *partially U-common* if $\sigma_k^{(r)} > 0$ for some blocks only;
- *U-specific* to block k if $\sigma_k^{(r)} > 0$ for this block only;
- *inactive* if $\sigma_k^{(r)} = 0$ for all k.

OMCOA always returns strictly positive weights, so that this pattern has to be assessed afterwards. We propose a Bayesian version of OMCOA in which a spike-and-slab prior on the weights $\sigma_k^{(r)}$ allows them to be exactly zero. The dimension R is then a *maximal* rank: inactive dimensions are switched off by the sampler, and the effective rank is inferred from the data.

The sparsity of the model is based on the diagonal terms $\sigma_k^{(r)}$ as follows

$$\sigma_k^{(r)} = \gamma_k^{(r)}\, s_k^{(r)}, \qquad \gamma_k^{(r)} \in \{0,1\}, \quad s_k^{(r)} > 0. \tag{5}$$

Priors.

- spike-and-slab on the weights: $\gamma_k^{(r)} \mid \theta_r \sim \text{Bernoulli}(\theta_r)$ and $s_k^{(r)} \sim \mathcal{N}^+(0, \lambda^{-1})$, i.e.

$$p(\sigma_k^{(r)} \mid \theta_r) = (1-\theta_r)\,\delta_0(\sigma_k^{(r)}) + \theta_r\,\mathcal{N}^+(\sigma_k^{(r)}; 0, \lambda^{-1});$$

- $\theta_r \sim \text{Beta}(a_\theta, b_\theta)$, one inclusion probability per dimension, shared by the K blocks.

Sharing θ_r across blocks borrows strength: a dimension that is active in several blocks is more likely to be active in another one, and a dimension that is inactive everywhere sees θ_r shrink towards zero.

As in the non-sparse case, the orthonormality of $\mathbf{U}$ and $\mathbf{V}_k$ gives

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\sum_{r=1}^{R}\sigma_k^{(r)}d_k^{(r)} + \sum_{r=1}^{R}(\sigma_k^{(r)})^2, \qquad d_k^{(r)} = \mathbf{u}^{(r)\intercal}\mathbf{X}_k\mathbf{v}_k^{(r)}. \tag{6}$$

Proposition 2 (Full conditionals)**.** *Let* $\mathbf{Q} \in \mathcal{M}_{N,N-1}(\mathbb{R})$ *be an orthonormal basis of* $\mathbf{1}_N^{\perp}$ *and write* $\mathbf{U} = \mathbf{Q}\tilde{\mathbf{U}}$*.*

1. ***Common space:*** $\tilde{\mathbf{U}} \mid \cdot \sim \mathrm{vMF}(\mathbf{Q}^\intercal \mathbf{F}_{\mathbf{U}})$ *with* $\mathbf{F}_{\mathbf{U}} = \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k$*.*

2. ***Block loadings:*** $\mathbf{V}_k \mid \cdot \sim \mathrm{vMF}(\mathbf{F}_{\mathbf{V}_k})$ *with* $\mathbf{F}_{\mathbf{V}_k} = \tau_k \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k$*. When* $\sigma_k^{(r)} = 0$*, the* r*-th column of* $\mathbf{F}_{\mathbf{V}_k}$ *is zero and* $\mathbf{v}_k^{(r)}$ *is uniform on the orthogonal complement of the other columns.*

3. ***Weights (joint update of*** $\gamma_k^{(r)}$ ***and*** $\sigma_k^{(r)}$***).*** *Let*

$$\omega_k = \tau_k + \lambda, \qquad m_k^{(r)} = \frac{\tau_k \, d_k^{(r)}}{\omega_k}.$$

   Then, with $\sigma_k^{(r)}$ *integrated out,*

$$P(\gamma_k^{(r)} = 1 \mid \cdot) = \frac{\theta_r B_k^{(r)}}{\theta_r B_k^{(r)} + 1 - \theta_r}, \qquad B_k^{(r)} = 2\sqrt{\frac{\lambda}{\omega_k}} \, \exp\Big(\frac{\omega_k (m_k^{(r)})^2}{2}\Big) \, \Phi\Big(m_k^{(r)} \sqrt{\omega_k}\Big),$$

   where Φ *is the standard normal c.d.f.; and*

$$\sigma_k^{(r)} \mid \gamma_k^{(r)} = 1, \cdot \;\sim\; \mathcal{N}^+(m_k^{(r)}, \omega_k^{-1}), \qquad \sigma_k^{(r)} \mid \gamma_k^{(r)} = 0, \cdot \;=\; 0.$$

4. ***Inclusion probabilities:*** $\theta_r \mid \cdot \sim \mathrm{Beta}(a_\theta + \sum_k \gamma_k^{(r)},\; b_\theta + K - \sum_k \gamma_k^{(r)})$*.*

5. ***Noise precisions:*** $\tau_k \mid \cdot \sim \mathrm{Gamma}(a_0 + (N-1)p_k/2,\; b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\intercal\|^2/2)$*.*

Sketch of proof. Items 1, 2, 4 and 5 follow as in the non-sparse sampler (linearity of (4) in $\mathbf{U}$ and $\mathbf{V}_k$, and conjugacy). For item 3, the terms of the log-posterior involving $\sigma_k^{(r)}$ are, by (4), $\tau_k d_k^{(r)} \sigma_k^{(r)} - \frac{\tau_k}{2}(\sigma_k^{(r)})^2$. Under $\gamma_k^{(r)} = 0$, $\sigma_k^{(r)} = 0$ and this quantity is 0. Under $\gamma_k^{(r)} = 1$, integrating against the half-normal slab of density $2\sqrt{\lambda/(2\pi)}\exp(-\lambda s^2/2)$ on $(0,\infty)$ gives

$$2\sqrt{\frac{\lambda}{2\pi}} \int_0^\infty \exp\Big(-\frac{\omega_k}{2} s^2 + \tau_k d_k^{(r)} s\Big) ds = 2\sqrt{\frac{\lambda}{\omega_k}} \exp\Big(\frac{\omega_k (m_k^{(r)})^2}{2}\Big) \Phi(m_k^{(r)} \sqrt{\omega_k}) = B_k^{(r)},$$

which is the Bayes factor of $\gamma_k^{(r)} = 1$ against $\gamma_k^{(r)} = 0$. The conditional of $\sigma_k^{(r)}$ given $\gamma_k^{(r)} = 1$ is obtained by completing the square. □

Updating $(\gamma_k^{(r)}, \sigma_k^{(r)})$ jointly, with $\sigma_k^{(r)}$ integrated out when drawing $\gamma_k^{(r)}$, is essential: drawing $\gamma_k^{(r)}$ given the current $\sigma_k^{(r)}$ would never allow the chain to leave $\sigma_k^{(r)} = 0$. In practice $\log B_k^{(r)}$ is computed on the log scale, with $\log \Phi$ evaluated directly, to avoid overflow.

4 Algorithm

The matrix von Mises–Fisher distributions are sampled by one column-wise sweep per iteration, each column being drawn from a vector von Mises–Fisher distribution with Wood's algorithm, as in the non-sparse sampler.

Algorithm 2 Sparse Gibbs sampler for OMCOA

Require: centered blocks $\mathbf{X}_1, \dots, \mathbf{X}_K$; maximal rank $R \le \min_k p_k$; iterations T; burn-in T_0; hyperparameters $a_0, b_0, \lambda, a_\theta, b_\theta$

Initialization: $\mathbf{U}, \mathbf{V}_k, \Sigma_k$ from OMCOA with R dimensions; $\gamma_k^{(r)} \leftarrow 1$; $\theta_r \leftarrow 1/2$; $\tau_k \leftarrow (N-1)p_k / \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2$
for $t = 1, \dots, T$ **do**
  update $\tilde{\mathbf{U}}$ by one sweep of $\mathrm{vMF}\big(\mathbf{Q}^\intercal \sum_k \tau_k \mathbf{X}_k \mathbf{V}_k \Sigma_k\big)$; $\mathbf{U} \leftarrow \mathbf{Q}\tilde{\mathbf{U}}$
  for $k = 1, \dots, K$ **do**
    update $\mathbf{V}_k$ by one sweep of $\mathrm{vMF}\big(\tau_k \mathbf{X}_k^\intercal \mathbf{U} \Sigma_k\big)$
    for $r = 1, \dots, R$ **do**
      $d_k^{(r)} \leftarrow \mathbf{u}^{(r)\intercal} \mathbf{X}_k \mathbf{v}_k^{(r)}$, $\omega_k \leftarrow \tau_k + \lambda$, $m_k^{(r)} \leftarrow \tau_k d_k^{(r)} / \omega_k$
      draw $\gamma_k^{(r)} \sim \mathrm{Bernoulli}\Big(\dfrac{\theta_r B_k^{(r)}}{\theta_r B_k^{(r)} + 1 - \theta_r}\Big)$
      if $\gamma_k^{(r)} = 1$ **then** draw $\sigma_k^{(r)} \sim \mathcal{N}^+(m_k^{(r)}, \omega_k^{-1})$ **else** $\sigma_k^{(r)} \leftarrow 0$
    end for
    draw $\tau_k \sim \mathrm{Gamma}\big(a_0 + (N-1)p_k/2,\ b_0 + \|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2/2\big)$
  end for
  for $r = 1, \dots, R$ **do**
    draw $\theta_r \sim \mathrm{Beta}\big(a_\theta + \sum_k \gamma_k^{(r)},\ b_\theta + K - \sum_k \gamma_k^{(r)}\big)$
  end for
  if $t > T_0$ **then** store $\big(\mathbf{U}, \Sigma_1, \dots, \Sigma_K, \boldsymbol{\gamma}, \boldsymbol{\theta}, \boldsymbol{\tau}\big)$
  end if
end for
align the stored draws (permutation and signs) on a reference $\mathbf{U}_{\mathrm{ref}}$
return the aligned draws

5 Posterior summaries

After alignment of the $T - T_0$ stored draws (indexed by t) on a reference solution, as in the non-sparse sampler:

- **Posterior inclusion probabilities.** $\hat{P}_k^{(r)} = \frac{1}{T-T_0} \sum_t \gamma_k^{(r),t}$ estimates $P(\gamma_k^{(r)} = 1 \mid \mathbf{X}_1, \dots, \mathbf{X}_K)$. The $K \times R$ matrix $(\hat{P}_k^{(r)})$ summarizes the structure of the blocks; a dimension is declared active in block k when $\hat{P}_k^{(r)} > 1/2$ (median probability model).
- **Effective rank.** $R_{\mathrm{eff}} = \#\{r : \sum_k \gamma_k^{(r)} > 0\}$, whose posterior distribution is estimated by its empirical distribution over the draws.
- **Block weights.** Posterior means and credible intervals of $\sigma_k^{(r)}$; for active entries they are computed conditionally on $\gamma_k^{(r)} = 1$.

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Appendix: proof of the proposition

Proof. The criterion to be minimized is:

$$\mathcal{L}(\mathbf{U}, \mathbf{V}_1, \ldots, \mathbf{V}_K, \Sigma_1, \ldots, \Sigma_K) = \sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\Sigma_k \mathbf{V}_k^\top\|^2,$$

subject to $\mathbf{U}^\intercal\mathbf{U} = I_R$ and $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$ for all $k = 1, \ldots, K$.

The algorithm is of the *Alternating Least Squares* (ALS) type: at each iteration, the criterion is minimized successively with respect to each parameter, while the others are held fixed. It therefore suffices to show that each individual update decreases or maintains $\mathcal{L}$.

Update of U With $\mathbf{V}_k$ and Σ_k fixed, let $\mathbf{H}_k = \Sigma_k \mathbf{V}_k^\intercal \in \mathcal{M}_{R,p_k}(\mathbb{R})$ and $\mathbf{H} = [\mathbf{H}_1 \mid \cdots \mid \mathbf{H}_K] \in \mathcal{M}_{R,P}(\mathbb{R})$. Then:

$$\mathcal{L} = \sum_{k=1}^{K} \|\mathbf{X}_k - \mathbf{U}\mathbf{H}_k\|^2 = \|\mathbf{X} - \mathbf{U}\mathbf{H}^\intercal\|^2.$$

Expanding: $\|\mathbf{X} - \mathbf{U}\mathbf{H}^\intercal\|^2 = \|\mathbf{X}\|^2 - 2\,\mathrm{trace}(\mathbf{X}^\intercal\mathbf{U}\mathbf{H}^\intercal) + \|\mathbf{U}\mathbf{H}^\intercal\|^2$.

Since $\mathbf{U}^\intercal\mathbf{U} = I_R$: $\|\mathbf{U}\mathbf{H}^\intercal\|^2 = \mathrm{trace}(\mathbf{H}\mathbf{U}^\intercal\mathbf{U}\mathbf{H}^\intercal) = \mathrm{trace}(\mathbf{H}\mathbf{H}^\intercal) = \|\mathbf{H}\|^2$ (constant w.r.t. $\mathbf{U}$).

Minimizing $\mathcal{L}$ with respect to $\mathbf{U}$ is therefore equivalent to maximizing

$$\mathrm{trace}(\mathbf{X}^\intercal\mathbf{U}\mathbf{H}^\intercal) = \mathrm{trace}(\mathbf{H}^\intercal\mathbf{X}^\intercal\mathbf{U}) = \mathrm{trace}(\mathbf{U}^\intercal\mathbf{X}\mathbf{H}),$$

subject to $\mathbf{U}^\intercal\mathbf{U} = I_R$. The first equality comes from $\mathrm{trace}(\mathbf{AB}) = \mathrm{trace}(\mathbf{AB})$ for $\mathbf{A}, \mathbf{B}$ of dimension N, P and P, N. The second equality comes from the equality of the trace of a square matrix and its transpose. By [Cliff, 1966], the maximum is attained at $\mathbf{U}^* = AB^\intercal$, where A (resp. B) are the matrices of the R leading left (resp. right) singular vectors of $\mathbf{X}\mathbf{H}^T$. subject to $\mathbf{U}^\intercal\mathbf{U} = I_R$. The first equality comes from $\mathrm{trace}(\mathbf{AB}) = \mathrm{trace}(\mathbf{AB})$ for $\mathbf{A}, \mathbf{B}$ of dimension N, P and P, N. The second equality comes from the equality of the trace of a square matrix and its transpose. By [Cliff, 1966], the maximum is attained at $\mathbf{U}^* = AB^\intercal$, where A (resp. B) are the matrices of the R leading left (resp. right) singular vectors of $\mathbf{X}\mathbf{H}^T$. This update is the global minimizer of $\mathcal{L}$ with respect to $\mathbf{U}$, hence:

$$\mathcal{L}(\mathbf{U}^*, \ldots) \leq \mathcal{L}(\mathbf{U}_{\text{old}}, \ldots).$$

Update of $\mathbf{V}_k$ With $\mathbf{U}$ and Σ_k fixed, expanding the contribution of block k:

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\ \mathrm{trace}(\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) + \|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2.$$

Since $\mathbf{U}^\intercal\mathbf{U} = I_R$ and $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$:

$$\|\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \mathrm{trace}(\mathbf{V}_k\Sigma_k^\intercal\mathbf{U}^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) = \mathrm{trace}(\Sigma_k^2) \quad \text{(constant w.r.t. } \mathbf{V}_k\text{)}.$$

Minimizing therefore reduces to maximizing $\mathrm{trace}(\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k\mathbf{V}_k^\intercal) = \mathrm{trace}(\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k)$ subject to $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$. Since the diagonal matrix $\Sigma_k > 0$ does not affect the singular vectors, the maximum is attained at $\mathbf{V}_k^* = \mathbf{A}\mathbf{B}^\intercal$ where $\mathbf{A}$, $\mathbf{B}$ are obtained from the SVD of $\mathbf{X}_k^\intercal\mathbf{U}$ [Cliff, 1966] . Hence: subject to $\mathbf{V}_k^\intercal\mathbf{V}_k = I_R$. Since the diagonal matrix $\Sigma_k > 0$ does not affect the singular vectors, the maximum is attained at $\mathbf{V}_k^* = \mathbf{A}\mathbf{B}^\intercal$ where $\mathbf{A}$, $\mathbf{B}$ are obtained from the SVD of $\mathbf{X}_k^\intercal\mathbf{U}$ [Cliff, 1966] . Hence:

$$\mathcal{L}(\ldots, \mathbf{V}_k^*, \ldots) \le \mathcal{L}(\ldots, \mathbf{V}_{k,\mathrm{old}}, \ldots).$$

Update of Σ_k With $\mathbf{U}$ and $\mathbf{V}_k$ fixed (both having orthonormal columns), and $\Sigma_k = \mathrm{diag}(\sigma_k^{(1)}, \ldots, \sigma_k^{(R)})$, the contribution of block k is:

$$\|\mathbf{X}_k - \mathbf{U}\Sigma_k\mathbf{V}_k^\intercal\|^2 = \|\mathbf{X}_k\|^2 - 2\ \mathrm{trace}(\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}\Sigma_k) + \mathrm{trace}(\Sigma_k^2).$$

Since Σ_k is diagonal, it remains to minimazing:

$$\sum_{r=1}^{R}\Big[()\sigma_k^{(r)})^2 - 2\,(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr}\cdot d_r\Big].$$

Each term is a convex parabola in $\sigma_k^{(r)}$, minimized at: $(\sigma_k^{(r)})^* = (\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)_{rr}$, or, in matrix form:$\Sigma_k^* = \mathrm{diag}(\mathbf{U}^\intercal\mathbf{X}_k\mathbf{V}_k)$. This update is therefore the exact minimizer with respect to Σ_k, and:

$$\mathcal{L}(\ldots, \Sigma_k^*, \ldots) \le \mathcal{L}(\ldots, \Sigma_{k,\mathrm{old}}, \ldots).$$

Global Convergence At each complete iteration (update of $\mathbf{U}$, then all $\mathbf{V}_k$, then all Σ_k), the criterion $\mathcal{L}$ is monotonically decreasing and bounded below by zero. By the monotone convergence theorem, the sequence of values $(\mathcal{L}^{(t)})_{t\ge 0}$ converges:

$$\mathcal{L}^{(t+1)} \le \mathcal{L}^{(t)}, \quad \mathcal{L}^{(t)} \ge 0 \implies \mathcal{L}^{(t)} \xrightarrow[t\to\infty]{} \ell^* \ge 0.$$

□

Sketch of proof. (1)–(2): by (4), the conditional density of $\mathbf{U}$ is proportional to $\exp\{\sum_k \tau_k\,\mathrm{trace}(\Sigma_k\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U})\} = \exp\{\mathrm{trace}(\mathbf{F}_\mathbf{U}^\intercal\mathbf{U})\}$, and $\mathrm{trace}(\mathbf{F}_\mathbf{U}^\intercal\mathbf{Q}\tilde{\mathbf{U}}) = \mathrm{trace}\big((\mathbf{Q}^\intercal\mathbf{F}_\mathbf{U})^\intercal\tilde{\mathbf{U}}\big)$; the same argument applies to $\mathbf{V}_k$ with $\mathrm{trace}(\Sigma_k\mathbf{V}_k^\intercal\mathbf{X}_k^\intercal\mathbf{U}) = \mathrm{trace}(\mathbf{F}_{\mathbf{V}_k}^\intercal\mathbf{V}_k)/\tau_k$. (3): the terms of (4) involving $\sigma_k^{(h)}$ are $-\frac{\tau_k}{2}\big[(\sigma_k^{(h)})^2 - 2\sigma_k^{(h)}d_k^{(h)}\big]$; adding the log-prior $-\frac{\lambda}{2}(\sigma_k^{(h)})^2$ and completing the square gives a Gaussian kernel, restricted to $\sigma_k^{(h)} > 0$. (4): the Gamma prior is conjugate for the precision of a Gaussian likelihood. □

Sampling from the matrix von Mises–Fisher distribution Direct sampling from vMF($\mathbf{F}$) is not straightforward, but its column-wise conditionals are vector von Mises–Fisher distributions [Hoff, 2009]. Let $\mathbf{W}_{-j}$ be $\mathbf{W}$ without its j-th column and $\mathbf{N}_j \in \mathcal{M}_{m,m-H+1}(\mathbb{R})$ an orthonormal basis of $\operatorname{span}(\mathbf{W}_{-j})^{\perp}$. Then

$$\mathbf{w}_j = \mathbf{N}_j \mathbf{z}, \qquad \mathbf{z} \sim \mathrm{vMF}\big(\mathbf{N}_j^{\intercal} \mathbf{f}_j\big) \text{ on the unit sphere } \mathbb{S}^{m-H},$$

where $\mathbf{f}_j$ is the j-th column of $\mathbf{F}$. One sweep over the H columns (in random order) is a valid Gibbs update of $\mathbf{W}$, which is enough inside the Gibbs sampler. The vector distribution vMF($\mathbf{f}$), with density proportional to $\exp(\mathbf{f}^{\intercal}\mathbf{z})$ on $\mathbb{S}^{m-1}$, is sampled with the rejection algorithm of Wood [Wood, 1994] (Algorithm 3).

Algorithm 3 Sampling $\mathbf{z} \sim \mathrm{vMF}(\mathbf{f})$ on $\mathbb{S}^{m-1}$ (Wood, 1994)

$\kappa \leftarrow \|\mathbf{f}\|, \quad \boldsymbol{\mu} \leftarrow \mathbf{f}/\kappa$
$b \leftarrow (m-1)/\big(2\kappa + \sqrt{4\kappa^2 + (m-1)^2}\big), \quad x_0 \leftarrow (1-b)/(1+b), \quad c \leftarrow \kappa x_0 + (m-1)\log(1-x_0^2)$
repeat
    draw $\zeta \sim \mathrm{Beta}\big(\frac{m-1}{2}, \frac{m-1}{2}\big)$ and $\upsilon \sim \mathcal{U}(0,1)$
    $w \leftarrow \big(1-(1+b)\zeta\big)/\big(1-(1-b)\zeta\big)$
until $\kappa w + (m-1)\log(1-x_0 w) - c \geq \log \upsilon$
draw $\boldsymbol{\xi} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_m)$, set $\boldsymbol{\xi} \leftarrow \boldsymbol{\xi} - (\boldsymbol{\mu}^{\intercal}\boldsymbol{\xi})\boldsymbol{\mu}$ and $\boldsymbol{\xi} \leftarrow \boldsymbol{\xi}/\|\boldsymbol{\xi}\|$
return $\mathbf{z} = w\,\boldsymbol{\mu} + \sqrt{1-w^2}\,\boldsymbol{\xi}$

The expression of b used in Algorithm 3 is algebraically equivalent to Wood's original one, $b = (-2\kappa + \sqrt{4\kappa^2 + (m-1)^2})/(m-1)$, but avoids a numerical cancellation when κ is large, which is the typical situation here since κ scales with the noise precisions τ_k.